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Nkululeko 1.0: A Python package to predict speaker characteristics with a high-level interface

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Software

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Summary

Nkululeko (Burkhardt, Wagner, et al., 2022) is open-source software written in Python and hosted on GitHub. It is predominantly a framework for audio-based machine learning explorations without the need to write Python code, and is strongly based on machine learning packages like sklearn (Pedregosa et al., 2011) and pytorch (Chaudhary et al., 2020). The main features are: training and evaluation of labelled speech databases with state-of-the-art machine learning approach and acoustic feature extractors, a live demonstration interface, and the possibility to store databases with predicted labels. Based on this, the framework can also be used to check on bias in databases by exploring correlations of target labels, like, e.g. depression or diagnosis, with predicted, or additionally given, labels like age, gender, signal distortion ratio or mean opinion score.

Design choices

The program is intended for **novice** people interested in speaker characteristics detection (e.g., emotion, age, and gender) without being proficient in (Python) programming language. Its main target is for **education** and **research** with the main features as follows:

- Finding good combinations of variables, e.g., acoustic features, models (classifier or regressor), feature standardization, augmentation, etc., for speaker characteristics detection (e.g., emotion);
- Characteristics of the database, such as distribution of gender, age, emotion, duration, data size, and so on with their visualization;
- Inference of speaker characteristics from a given audio file or streaming audio (can be said also as "weak" labeling for semi-supervised learning).

Hence, one should be able to use Nkululeko after installing and preparing/downloading their data in the correct format in a single line.

```
$ nkululeko.MODULE_NAME --config CONFIG_FILE.ini
```

How does it work?

nkululeko is a command line tool written in Python, best used in conjunction with the Visual Studio Code editor (but can be run stand-alone). To use it, a text editor is needed to edit the experiment configuration. You would then run nkululeko like this:

```
$ nkululeko.explore --config conf.ini
```

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and inspect the results afterward; they are represented as images, texts, and even a fully automatically compiled PDF report written in latex.

nkululeko's data import format is based on a simple CSV formalism, or alternatively, for a more detailed representation including data schemata, audformat.[1] Basically, to be used by nkululeko, the data format should include the audio file path and a task-specific label. Optionally, speaker ID and gender labels help with speech data. An example of a database labelled with emotion is

```
file, speaker, gender, emotion
x/sample.wav, s1, female, happy
...
```

As the main goal of nkululeko is to avoid the need to learn programming, experiments are specified by means of a configuration file. The functionality is encapsulated by software *modules* (interfaces) that are to be called on the command line. We list the most important ones here:

- **nkululeko**: do machine learning experiments combining features and learners
- **demo**: demo the current best model on the command line or some files
- **test**: run the current best model on a specified test set
- **explore**: perform data exploration (used mainly in this paper)
- **augment**: augment the current training data. This could also be used to reduce bias in the data, for example, by adding noise to audio samples that belong to a specific category.
- **aug_train**: augment the training data and train the model with the augmented data.
- **predict**: predict features like speaker diarization, signal distortion ratio, mean opinion score, arousal/valence, age/gender (for databases that miss this information), with deep neural nets models, e.g. as a basis for the *explore* module.
- **segment**: segment a database based on VAD (voice activity detection)
- **ensemble**: ensemble several models to improve performance

The configuration (INI) file consists of a set of key-value pairs that are organised into several sections. Almost all keys have default values, so they do not have to be specified.

Here is a sample listing of an INI file (conf.ini) with a database section:

```
[EXP]
name = explore-androids
[DATA]
databases = ['androids']
androids = /data/androids/androids.csv
target = depression
labels = ['depressed', 'control']
samples_per_speaker = 20
min_length = 2
[PREDICT]
sample_selection = all
targets = ['pesq', 'sdr', 'stoi', 'mos']
[EXPL]
value_counts = [['gender'], ['age'], ['est_sdr'], ['est_pesq'], ['est_mos']]
[REPORT]
latex = androids-report
```

As can be seen, some of the values simply contain Python data structures like arrays or dictionaries. Within this example, an experiment is specified with the name *explore-androids*,

[1] https://audeering.github.io/audformat/

and a result folder with this name will be created, containing all figures and textual results, including an automatically generated Latex and PDF report on the findings.

The *DATA* section sets the location of the database and specifies filters on the sample, in this case limiting the data to 20 samples per speaker at most and at least 2 seconds long. In this section, the split sets (training, development, and test) are also specified. There is a special feature named *balance splits* that lets the user specify criteria that should be used to stratify the splits, for example, based on signal distortion ratio.

With the *predict* module, specific features like, for example, signal distortion ratio or mean opinion score are to be predicted by deep learning models. The results are then used by a following call to the *explore* module to check whether these features, as well as some ground truth features (*age* and *gender*), correlate with the target variable (*depressed* in the given example) in any way.

The `nkululeko` configuration can specify further sections:

- **FEATS** to specify acoustic features (e.g. opensmile (Eyben et al., 2010) or deep learning embeddings; e.g. wav2vec 2.0 (Baevski et al., 2020)) that should be used to represent the audio files.
- **MODEL** to specify statistical models for regression or classification of audio data.

Example of usage

In the previous section, we have seen how to specify an experiment in an INI file that can be run with, for instance, `explore` and `segment` modules. Here, we show how to run the experiment (`nkululeko.nkululeko`) with built-in dataset (Polish Speech Emotions dataset) from the installation until getting the results.

First, novices could clone the GitHub repository of nkululeko.

```
$ git clone https://github.com/felixbur/nkululeko.git
$ cd nkululeko
```

Then, install nkululeko with `pip`. It is recommended that a virtual environment be used to avoid conflicts with other Python packages.

```
$ python -m venv .env
$ source .env/bin/activate
$ pip install nkululeko
```

Next, extract `polish_speech_emotions.zip` inside the nkululeko data folder (`nkululeko/data/polish`) with right click regardless of the operating system (or using `unzip` command in the terminal like below). Then, run the following command in the terminal:

```
$ cd data/polish
$ unzip polish_speech_emotions.zip
$ python3 process_database.py
$ cd ../..
$ nkululeko.nkululeko --config data/polish/exp.ini
```

That's it! The results will be stored in the `results/exp_polish_os` folder as stated in `exp.ini`. Below is an example of the debug output of the command:

```
DEBUG: nkululeko: running exp_polish_os from config data/polish/exp.ini,
nkululeko version 0.91.0
...
DEBUG: reporter:
               precision    recall  f1-score   support
```

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```
       anger     0.6944    0.8333    0.7576        30
     neutral     0.5000    0.4333    0.4643        30
        fear     0.6429    0.6000    0.6207        30

    accuracy                         0.6222        90
   macro avg     0.6124    0.6222    0.6142        90
weighted avg     0.6124    0.6222    0.6142        90

DEBUG: reporter: labels: ['anger', 'neutral', 'fear']
DEBUG: reporter: result per class (F1 score): [0.758, 0.464, 0.621]
from epoch: 0
DEBUG: experiment: Done, used 7.439 seconds
DONE
```

What has been added since the last publication

Besides many small changes, mainly three big additions extended nkululeko's functionality since the last published publications. We introduce them in the next subsections.

Finetune transformer models

With nkululeko since version 0.85.0 you can finetune a transformer model with huggingface (and even publish it there if you like).

Finetuning in this context means to train the (pre-trained) transformer layers with your new training data labels, as opposed to only using the last layer as embeddings.

The only thing you need to do is to set your MODEL type to *finetune*:

```
[FEATS]
type = []
[MODEL]
type = finetune
```

The acoustic features can/should be empty, because the transformer model starts with CNN layers to model the acoustics frame-wise. The frames are then getting pooled by the model for the whole utterance.

The default base model is the one from facebook, but you can specify a different one like this:

```
[MODEL]
type = finetune
pretrained_model = microsoft/wavlm-base

duration = 10.5
```

The parameter *max_duration* is also optional (default=8) and means the maximum duration of your samples / segments (in seconds) that will be used, starting from 0. The rest is disregarded.

You can use the usual deep learning parameters:

```
[MODEL]
learning_rate = .001
batch_size = 16
device = cuda:3
measure = mse
loss = mse
```

but all of them have defaults.

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The loss function is fixed to

- weighted cross entropy for classification
- concordance correlation coefficient for regression

The resulting best model and the huggingface logs (which can be read by tensorboard) are stored in the project folder.

If you like to have your model published, set:

```
[MODEL]
push_to_hub = True
```

Ensemble classification

With nkululeko since version 0.88.0 you can combine experiment results and report on the outcome, by using the **ensemble** module.

For example, you would like to know if the combination of expert features and learned embeddings works better than one of those. You could then do

```
python -m nkululeko.ensemble \
--method max_class \
tests/exp_emodb_praat_xgb.ini \
tests/exp_emodb_ast_xgb.ini \
tests/exp_emodb_wav2vec_xgb.in
```

(all in one line) and would then get the results for a majority voting of the three results for Praat, AST and Wav2vec2 features.

Other methods to combine the different predictors, are *mean*, *max*, *sum*, *max_class*, *uncertainty_threshold*, *uncertainty_weighted*, *confidence_weighted*:

- **majority_voting**: The modality function for classification: predict the category that most classifiers agree on.
- **mean**: For classification: compute the arithmetic mean of probabilities from all predictors for each labels, use highest probability to infer the label.
- **max**: For classification: use the maximum value of probabilities from all predictors for each labels, use highest probability to infer the label.
- **sum**: For classification: use the sum of probabilities from all predictors for each labels, use highest probability to infer the label.
- **max_class**: For classification: compare the highest probabilities of all models across classes (instead of same class as in max_ensemble) and return the highest probability and the class
- **uncertainty_threshold**: For classification: predict the class with the lowest uncertainty if lower than a threshold (default to 1.0, meaning no threshold), else calculate the mean of uncertainties for all models per class and predict the lowest.
- **uncertainty_weighted**: For classification: weigh each class with the inverse of its uncertainty (1/uncertainty), normalize the weights per model, then multiply each class model probability with their normalized weights and use the maximum one to infer the label.
- **confidence_weighted**: Weighted ensemble based on confidence (1-uncertainty), normalized for all samples per model. Like before, but use confidence (instead of inverse of uncertainty) as weights.

Predicting Speaker ID

To have labels for the individual speakers in a database is extremely important, because if you mix the same speakers in training and testing data splits, it is very possible that your model

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simply learned some speaker idiosyncrasies instead of some underlying principle. If you don't have this labels, you could at least try to infer them with a pre-trained model.

With nkululeko since version 0.93.0 the pyannote segmentation package is interfaced (as an alternative to silero)

There are two modules that you can use for this:

- SEGMENT
- PREDICT

The (huge) difference is, that the SEGMENT module looks at each file in the input data and looks for speakers per file (can be only one large file), while the PREDICT module concatenates all input data and looks for different speakers in the whole database.

In any case best run it on a GPU, as CPU will be very slow (and there is no progress bar).

If you specify the *method* in [SEGMENT] section and the *hf_token* (needed for the pyannote model) in the [MODEL] section

```
[SEGMENT]
method = pyannote
segment_target = _segmented
sample_selection = all
[MODEL]
hf_token = <my hugging face token>
```

your resulting segmentations will have predicted speaker id attachched.. Be aware that this is really slow on CPU, so best run on GPU and declare so in the [MODEL] section:

```
[MODEL]
hf_token = <my hugging face token>
device=gpu # or cuda:0
```

As a result a new plot would appear in the image folder: the distribution of speakers that were found.

Simply select *speaker* as the prediction target:

```
[PREDICT]
targets = ["speaker"]
```

Generally, the PREDICT module is described here

Statement of need

Open-source tools are believed to be one of the reasons for accelerated science and technology. They are more secure, easy to customise, and transparent. There are several open-source tools that exist for acoustic, sound, and audio analysis, such as librosa (McFee et al., 2015), TorchAudio (Yang et al., 2021), pyAudioAnalysis (Giannakopoulos, 2015), ESPNET (Watanabe et al., 2018), and SpeechBrain (Ravanelli et al., 2021). However, none of them are specialised in speech analysis with high-level interfaces for novices in the speech processing area.

One exception is Spotlight (Suwelack, 2023), an open-source tool that visualises metadata distributions in audio data. An existing interface between `nkululeko` and Spotlight can be used to combine the visualisations of Spotlight with the functionalities of Nkululeko.

Nkululeko follows these principles:

- *Minimum programming skills*: The only programming skills required are preparing the data in the correct (CSV) format and running the command line tool. For AUDFORMAT, no preparation is needed.

- *Standardised data format and label*: The data format is based on CSV and AUDFORMAT, which are widely used formats for data exchange. The standard headers are like 'file', 'speaker', 'emotion', 'age', and 'language' and can be customised. Data could be saved anywhere on the computer, but the recipe for the data preparation is advised to be saved in `nkululeko/data` folder (and/or make a soft link to the original data location).
- *Replicability*: the experiments are specified in a configuration file, which can be shared with others including the splitting of training, development, and test partition. All results are stored in a folder with the same name as the experiment.
- *High-level interface*: the user specifies the experiment in an INI file, which is a simple text file that can be edited with any text editor. The user does not need to write Python code for experiments.
- *Transparency*: as CLI, nkululeko *always output debug*, in which info, warning, and error will be obviously displayed in the terminal (and should be easily understood). The results are stored in the experiment folder for further investigations and are represented as images, texts, and even a fully automatically compiled PDF report written in latex.

Usage in existing research

Nkululeko has been used in several research projects since its first appearance in 2022 (Burkhardt, Wagner, et al., 2022). The following list gives an overview of the research papers that have used Nkululeko:

- (Burkhardt, Eyben, et al., 2022): this paper reported a database development of synthesized speech for basic emotions and its evaluation using the Nkululeko toolkit.
- (Burkhardt et al., 2024): this paper shows how to use Nkululeko for bias detection. The findings on two datasets, UACorpus and Androids, show that some features are correlated with the target label, e.g., depression, and can be used to detect bias in the database.
- (Atmaja et al., 2024): this paper shows Nkululeko's capability for ensemble learning with a focus on uncertainty estimation.
- (Atmaja & Sasou, 2025): in this paper, evaluations of different handcrafted acoustic features and SSL approaches for pathological voice detection tasks were reported, highlighting the ease of using Nkululeko to perform extensive experiments including combinations of different features at different levels (early and late fusions).
- (Atmaja et al., 2025): this paper extends the previous ensemble learning evaluations with performance weighting (using weighted and unweighted accuracies) on five tasks and ten datasets.

Changes

Nkululeko has been described in three papers so far, we give a short overview on the updates since then.

- **2022 Paper:** F. Burkhardt, Johannes Wagner, Hagen Wierstorf, Florian Eyben and Björn Schuller: Nkululeko: A Tool For Rapid Speaker Characteristics Detection, Proc. Proc. LREC, 2022. **New features:** First version mainly focussing on basic machinelearning experiments that combine *expert* acoustic features (like Praat or opensmile features) with traditional learning approaches.
- **2023 Paper:** F. Burkhardt, Florian Eyben and Björn Schuller: Nkululeko: Machine Learning Experiments on Speaker Characteristics Without Programming, Proc. Inter-

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speech, 2023. **New features:** Mainly extending the acoustic features to deep-learning based (like TRILL, Hubert or wav2vec2) and the models by neural net architectures like MLP or CNN.

- **2024 Paper:** F. Burkhardt, Bagus Tris Atmaja, Anna Derington, Florian Eyben and Björn Schuller: Check Your Audio Data: Nkululeko for Bias Detection, Proc. Oriental COCOSDA, 2024. **New features:** Introducing the concept of interfaces (or *modules*), focusing on the *explore-module* that features automatic data statistics and and bias analysis.
- **Since then:** Besides many minor enhancements; ensemble learning, Wav2vec2 model finetuning, adding automatic speaker identification, extending augmentation and segmentation.

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